

International Institute of Information Technology, Hyderabad

CE8.401 Disaster Management

(Spring 2026)

Mid Semester Examination

Max. Time: 1.5 Hrs

Max. Marks: 40

-
1. A town is located in Seismic Zone IV. A seismic hazard study indicates that the probability of experiencing at least one damaging earthquake in 50 years is 15%. The town has 1500 residential buildings, with an average cost of ₹50 lakh per building. The buildings have a vulnerability factor of 0.25. Based on the above data, calculate:
 - (a) The annual hazard rate (2)
 - (b) The total exposure (2)
 - (b) Risk in terms of Annual Loss due to earthquakes (3)
 - (d) The percentage reduction in risk if vulnerability is reduced to 0.10 by retrofitting (3)

 2. Suppose you are given about 100 TB of global disaster data consisting of satellite images, seismic time series, weather records, population data and infrastructure maps.
 - (a) What would be an appropriate prediction target for a disaster machine learning model based on this data? Clearly define the prediction target and justify why it would be useful for disaster management. (3)
 - (b) Explain why it is unlikely that a single machine learning model can be used effectively for predicting all types of disasters. (3)
 - (c) Propose a modelling framework that integrates physics-based understanding, data-driven models and infrastructure information. Explain how these components would work together to estimate disaster risk or impact. (4)
 - (d) Suppose a model predicts a major earthquake tomorrow with a probability of 60%. Should authorities issue a public warning based on this prediction? Justify your answer by discussing both technical considerations and societal implications. (4)

 3. State four advantages and four challenges associated with the retrofitting of damaged buildings. (4)

4. In October 2030, a coastal industrial city experienced a severe tropical cyclone during a period of unusually high tide. The cyclone brought intense rainfall and strong winds that caused widespread flooding in low-lying areas. The saturated soil conditions triggered several landslides along nearby hill slopes, blocking major highways and delaying emergency response.

Floodwaters entered a large chemical storage facility, causing leakage of hazardous materials into nearby water bodies. The contamination forced shutdown of the city's main water treatment plant. Due to flooding of electrical substations, power supply was interrupted across the city. The power outage caused failure of telecom networks and disruption of internet connectivity. Several manufacturing industries were forced to halt production, leading to economic losses and supply chain delays across the region.

(a) From the scenario above, identify one example each of:

- Compound risk
- Interacting risk
- Interconnected risk
- Cascading risk

Explain your reasoning clearly.

(4 marks)

(b) Construct a flow diagram showing the progression of the disaster from the initial triggering events to major societal impacts. The diagram should include at least 7 nodes.

(4 marks)

(c) Identify two escalation points in the system and explain why they increased the severity of the disaster.

(2 marks)

(d) Explain why this disaster cannot be described using only a single risk category.

(2 marks)